

Can Cryptocurrencies Realistically Perform the Three Traditional Functions of Money?

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Contents

Abstract 3

Introduction 5

The Traditional Functions of Money: A Theoretical Framework 6

Bitcoin as a Medium of Exchange: Architectural Limits to Scalability 7

- The Blockchain Trilemma and Consensus Mechanisms 7
- Transaction Fees and Congestion Dynamics 7
- Settlement Finality vs. Instant Clearance 8

Bitcoin as a Store of Value: The Paradox of Absolute Scarcity 8

- The Mechanics of Algorithmic Scarcity 8
- The Volatility Problem 9
- Bitcoin and Market Risk: A Pro Cyclical Asset 9

Bitcoin as a Unit of Account: Real Time Volatility and Fiat Dominance 10

- The Impact of Volatility on Corporate Accounting 10
- Dynamic Pricing and Fiat Asymmetry 10
- The El Salvador Natural Experiment 11

Stablecoins as a Comparison: Bridging the Monetary Gap 12

- Typologies of Stablecoins 12

Conclusion 14

References 16

Abstract

The Main Problem: Decentralized crypto assets appeared in the aftermath of the 2008 global financial crisis in an attempt to compete with sovereign fiat monetary systems. Cryptocurrencies were developed to replace money and the traditional banks. But when they got popular, there was a big difference between their theoretical goals and how they actually work. Cryptocurrencies are not real money, because they can't do three basic things at the same time, because of limits in their internal technology. They can't work as a simple way to pay, hold their value safely over time, and be used to set stable prices for goods and services. (Brunner and Meltzer, 1971).

Indication of Methodology: This paper is basically a secondary research which comprises data and information from different journals, article and sources from internet. It examines the design of blockchain systems closely. It uses the rules of traditional monetary theory to compare older, slow crypto networks, such as Bitcoin, with newer financial technologies.

The Key Findings:

Older crypto networks are too slow, topping out at around 7 transactions per second, and their network fees are all over the place. This makes them completely useless for buying everyday stuff like coffee.

Second, the supply of Bitcoin is locked in by code, and cannot change as demand changes. So its price goes up and down violently. This extreme volatility makes it unusable for normal companies for normal accounting or payroll. (Yermack, 2015).

Third, newer “stablecoins” solve this price problem by adjusting their value directly to normal government money like the US dollar. However, this comes at the cost of reliance on centralized banks and custodians, thus losing the fundamental crypto dream of full independence (Gorton and Zhang, 2021).

Cryptocurrencies can't fully replace normal money at the moment. Instead, the ecosystem has divided into different tools: Bitcoin is used as “digital gold” to store wealth over the long term; stablecoins are used as digital rails for payments; and private blockchains are used by large banks to upgrade their internal plumbing. Traditional money is still heavily controlled by government systems.

Introduction

Bitcoin emerged in 2008 as a disruptive alternative to the current financial system: a decentralized peer to peer electronic cash system that is supposed to operate beyond the reach of central banks and traditional financial intermediaries. Developed in the aftermath of the 2008 global financial crisis, this technology aimed to replace institutional trust with cryptographic proof (Nakamoto 2008). Over the coming decades this basic framework has given rise to a multi trillion dollar digital asset ecosystem and reshaped how regulators, commercial banks and market participants view financial services. (Ballis and Drakos 2021).

But when the technology moved from being a cypherpunk experiment to becoming a mainstream asset class, there was a huge gap between what it aspired to be in theory and what it actually was in practice. But enthusiasts believe that decentralized networks are the expected next stage in the evolution of currency, while traditional monetary economists disagree. This is not an academic debate, it is at the heart of macroeconomics and financial stability. To determine if cryptocurrencies can realistically supplant or complement fiat currencies, they must be critically assessed against the three classical functions of money – medium of exchange, store of value, and unit of account. This report shows a thorough, evidence based look at blockchain architecture. It explores why native crypto assets like Bitcoin have a hard time meeting these functions at the same time. It also evaluates how later FinTech innovations, like stablecoins, try to fix these ongoing issues.

The Traditional Functions of Money: A Theoretical Framework

This is a safe way of moving purchasing power from the present to the future. An asset that maintains its value over time should exhibit relative price stability, liquid secondary markets, and resistance to sudden shocks in supply or value. In monetary theory, an asset is recognized as money not because of its physical or digital form, but based on its usefulness in three interconnected areas (Brunner and Meltzer 1971)

• **Medium of Exchange:** This is an instrument used to facilitate transactions of goods and services. A good medium of exchange removes the "double coincidence of wants" found in barter systems (Mishkin, 2020). This significantly lowers transaction costs. For it to work globally, it needs to have low friction, high throughput, predictable costs, and nearly universal acceptance.

• **Store of Value:** It's a good way to transfer buying power from today to in future. For an asset to maintain value over time, it must be relatively stable in price, have secondary markets with high liquidity, and not subject to sudden drops in value or unpredictable changes in supply.

• **Unit of Account:** This is the standard measure used to price goods, services, assets, liabilities, and economic calculations in an economy. A trustworthy unit of account allows individuals and businesses to set prices for long term contracts, calculate profit margins, and make sound economic plans.

Importantly, these functions support each other. An asset cannot effectively serve as a unit of account if its ability to act as a store of value is highly unpredictable. Likewise, its usefulness as a medium of exchange is reduced if people cannot anticipate its value (Brunner and Meltzer 1971).

Bitcoin as a Medium of Exchange: Architectural Limits to Scalability

Bitcoin was designed to be a peer to peer digital currency. It avoids using centralized payment systems like Visa or SWIFT. However, when we look at its practical use as a global medium of exchange, the blockchain's structure reveals fixed barriers.

The Blockchain Trilemma and Consensus Mechanisms

The Blockchain Trilemma, which holds that a decentralized ledger can only satisfy two of three functions: decentralization, security, or scalability. These are the primary cause of Bitcoin's operational limitations. Bitcoin's Proof of Work (PoW) consensus method puts security and decentralization ahead of scalability. A worldwide network of nodes must come to an agreement by resolving resource intensive cryptographic puzzles in order to validate transactions without the need for a trusted third party (Nakamoto 2008).

• **Block Size and Generation Time:** Bitcoin strives for a block generation time of 10 minutes and restricts the size of its blocks to about 1 megabyte . (Antonopoulos, 2017, p. 143).

• **Transaction Limitations**: Consequently, the maximum throughout of the base layer network is essentially restricted to around 7 transactions per second (TPS). On the other hand, legacy centralized networks, such as Visa, frequently manage over 24,000 TPS and grow to even higher volumes during times of peak demand. (Okhaifo, 2026)

Transaction Fees and Congestion Dynamics

The network uses a competitive, market driven incentive structure to prioritize transactions since block space is severely limited. Users are forced to compete with one another by charging miners greater transaction fees when payment volume exceeds block availability. (Huberman et al.

2021). Transaction costs often rise to unaffordable levels during times of high market speculation or network congestion. Because of this price structure, Bitcoin is completely impractical for regular retail purchases or micro transactions. An economically illogical breakdown of a medium of trade is shown by paying a $15 network charge for a $4 cup of coffee.

Settlement Finality vs. Instant Clearance

Even if backend settlement takes days, a point of sale transaction in traditional banking seems immediate to the customer. On the other hand, Bitcoin needs on chain cryptographic settlement finality. A transaction is usually not deemed safe until it is checked behind at least six following blocks, which takes around 60 minutes, due to the possibility of “chain reorganization”. (Nakamoto, 2008). This structural delay renders base layer Bitcoin payments impractical for contemporary retail situations that need sub second clearing.

Bitcoin as a Store of Value: The Paradox of Absolute Scarcity

The story around Bitcoin changed dramatically as its practical constraints as a means of trade became apparent. More and more supporters refer to Bitcoin as "Digital Gold"—a top global store of value—rather than digital currency.

The Mechanics of Algorithmic Scarcity

The predetermined supply curve of Bitcoin is the foundation for its claim to be a store of value. Bitcoin has an absolute supply ceiling of 21 million coins, in contrast to fiat currencies, which central banks may inflate arbitrarily through monetary expansion and quantitative easing. An automatic process called the "halving," which reduces the issuance rate of fresh block rewards by 50% every 210,000 blocks—roughly every four years—enforces this scarcity (Nakamoto 2008).

Bitcoin provides a reliable hedge against long term fiat inflation and sovereign currency depreciation by eliminating human judgment from monetary supply. This structural censorship resistance offers a special means of capital preservation for investors working in hyperinflationary economies or unpredictable geopolitical regimes.

The Volatility Problem

However, an inelastic supply curve in Bitcoin's construction significantly compromises its dependability as a short to medium term store of wealth. A significant increase in demand in conventional commodity markets (such physical gold) prompts producers to intensify mining activities, boosting market supply and mitigating price shocks. The difficulty adjustment mechanism in Bitcoin's design guarantees that the pace of issuance stays constant regardless of how much demand or processing power enters the network. The price of the asset absorbs all of the volatility in demand since supply is unable to adapt to changing market conditions. (Cachanosky, 2019). As a result, the asset regularly experiences sharp, cyclical drawdowns, losing between 50% and 80% of its value in a matter of months. (BlackRock, 2024) An asset with such sharp price fluctuations does not meet the fundamental need of a reliable, stable store of value for institutional players, businesses handling payroll, or people looking to protect their immediate life savings.

Bitcoin and Market Risk: A Pro Cyclical Asset

Perhaps the most damaging evidence against Bitcoin as a store of value is its behaviour during market crises — just the times when a true store of value should do its job. The 60-day cyclical correlation of Bitcoin to the S&P 500 index peaked at 0.65 during the COVID-19 market crash in March 2020, and the monetary tightening cycle of 2022. Its price history also includes several drawdowns of over 75%, suggesting a serious failure to preserve capital for a supposed store of

value (Soleimani, 2025). Gold, on the other hand, has traditionally maintained or increased its value in times of financial stress, which is why it has held its status as the world's primary monetary store of value for centuries (Baur and Dimpfl, 2021). While bitcoin's correlation to stocks showed some signs of weakness, it remained robust during risk off episodes of market stress or swift deleveraging, demonstrating how institutional adoption has significantly diminished bitcoin's idiosyncratic behaviour. (Wu et al. 2025). This pro cyclical crisis behavior is in deep contradiction to the store of value story.

Bitcoin as a Unit of Account: Real Time Volatility and Fiat Dominance

The unit of account, the third pillar of monetary functionality, is the biggest operational flaw in Bitcoin. An asset's worth in relation to actual commodities and services must be steady over extended periods of time in order for it to be a useful yardstick for economic calculations.

The Impact of Volatility on Corporate Accounting

It is nearly hard for enterprises to use Bitcoin as a native accounting denomination due to its sharp hourly fluctuations in purchasing power. A manufacturing company's whole financial sheet would be exposed to catastrophic currency risk if it priced its long term corporate supplier contracts or staff compensation directly in Bitcoin. For example, a company's profit margins may be suddenly destroyed by a 20% drop in the price of Bitcoin over the course of a weekend, making it impossible for it to pay fixed liabilities that are still denominated in local fiat currency.

Dynamic Pricing and Fiat Asymmetry

Businesses seldom accept payment in Bitcoin when they do accept it at the time of sale, instead things are priced in a local fiat currency (e.g., USD, GBP, or EUR), which is instantly converted

into an equal fraction of Bitcoin using a dynamic checkout API. Bitcoin is only a highly volatile speculative vehicle used as a temporary payment rail in this transaction flow; it is not a unit of account. The fiat monetary system continues to be the foundation of the underlying economic calculation.

The El Salvador Natural Experiment

The primary evidence shows that bitcoin is not a suitable unit of account comes from El Salvador, the only country to have made it legal. Big government incentives, including a $30 sign up bonus through the Chivo Wallet app, did little to expand Bitcoin beyond one rich demographic: young, educated men with bank accounts. (Alvarez, Van Patten, & Argente, 2023). By 2022 Bitcoin's price was down 66%, and a 2022 study found 86% of local firms had no Bitcoin transactions. Firms were especially hesitant to adopt Bitcoin, fearing that the cryptocurrency's wild swings would be bad for business. In February 2025, El Salvador completely revoked Bitcoin's legal tender status, the last real-world failure of Bitcoin as a functional monetary unit of account.

Stablecoins as a Comparison: Bridging the Monetary Gap

The FinTech industry created stablecoins, a subclass of digital assets, after realizing that native cryptocurrencies cannot perform the fundamental functions of money due to their excessive price volatility. The tokens were created to combine the transactional efficiency of a blockchain ledger with the price stability of major fiat currencies.

Typologies of Stablecoins

- **Fiat Collateralized (Centralized):** Each currency is backed by a real dollar, stablecoins like Tether (USDT) and USD currency (USDC) remain stable. The companies keeps a dollar (or something similar to cash, such short term government bonds) in a bank reserve for each token that is created. To demonstrate that the backing is genuine, these reserves are audited.
- **Crypto Overcollateralized (Decentralized):** In order to produce an amount of stable tokens, assets like DAI, which are governed by smart contracts, require users to put up excess volatile cryptocurrency assets (such as 150% of Ethereum's value) as collateral. Automated liquidation mechanisms then absorb market shocks. (MakerDAO, 2017; Klages-Mundt and Minca, 2021).
- **Algorithmic (Uncollateralized):** These tokens respond to market demand by expanding or contracting their supply using programmable, game theoretic supply manipulation algorithms. As seen by the disastrous collapse of the Terra USD (USDT) ecosystem in 2022, this architecture proved to be very sensitive to speculative death spirals. (Clements, 2021; Cho, 2023)

Functional Dimension	Native Bitcoin (BTC)	Stablecoins (USDT/USDC)	Traditional Fiat	Decentralization Profile
Medium of Exchange	Low: Slow throughput, erratic fees, and volatility.	High: Instant settlement over public blockchains.	High: Mandated legal tender and pervasive banking rails.	Varies by asset type
Store of Value	Speculative: Strong long term scarcity, high short term risk.	Moderate: Preserves fiat value, but inherits inflation.	Moderate: Highly predictable short term, inflated long term.	Varies by asset type
Unit of Account	Failed: High asset volatility prevents native pricing.	High (Indirect): Effectively mirrors the fiat unit.	Absolute: Foundational benchmark for contracts and tax.	None (Centralized status)
System Governance	High: Trustless, global, censorship resistant network.	Low: Subservient to centralized custodians/freezes.	None: Managed entirely by monetary authorities.	Centralized vs Network nodes

Typically, stablecoins beat Bitcoin in the short term on the factors of medium of exchange and unit of account. Both, however, are not full blown types of money. In the end, stablecoins are parasites on the fiat monetary system they are tied to, and their stability relies on the integrity of reserves and governance procedures that have been shown to be ineffective. (Garratt & van Oordt, 2023). But bitcoin is still better described as a speculative asset than a useful currency. (Baur, et al., 2018; Yermack, 2015)

Conclusion

The adoption of cryptocurrencies has forever changed the course of contemporary finance, challenging long standing beliefs about the state monopoly on money. But an empirical evaluation of their structural architecture shows that native crypto assets fail to perform simultaneously the three traditional functions of money.

Bitcoin in practice has evolved away from its original goal of peer to peer electronic cash. Its Proof of Work architecture deliberately favors absolute decentralization and security over transactional throughput, resulting in serious scaling bottlenecks and variable fees that render it an unviable everyday medium of exchange. Furthermore, its supply curve is perfectly inelastic ensuring structural volatility in the price rendering it worthless as a stable unit of account. Rather, Bitcoin functions mostly as a macro level, speculative store of value. It is a type of digital gold that is sought after for its programmed scarcity and resilience to censorship, but it is not appropriate for everyday business calculations.

The FinTech sector created collateralized stablecoins to address this need. Stablecoins effectively address the volatility dilemma by tying their value to fiat money, providing a highly useful digital medium of exchange and an efficient unit of account across international networks. However, in order to achieve this stability, centralized custodial custody and regulatory compliance concerns are reintroduced, thereby replicating the exact fiat banking weaknesses that the blockchain movement first aimed to eliminate. In the end, cryptocurrencies are not yet a complete method of payment. Instead, the ecosystem has broken up into specialized digital tools: permissioned corporate blockchain ledgers improve back end institutional plumbing, stablecoins

function as digital payment rails, and native cryptocurrency assets serve as speculative sovereign hedging vehicles. The conventional uses of money will stay firmly rooted in the realm of the fiat monetary system until a decentralized network can create a stable asset that decouples from a fiat peg without sacrificing scalability or structural security.

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